

Region Growing and Region Splitting

Region Growing and Region Splitting

Edges and thresholds sometimes do not give good results for segmentation.

Region-based segmentation is based on the connectivity of similar pixels in a region.

There are two main approaches to region-based segmentation : region-growing and region-splitting.

Region Growing and Region Splitting

Procedure for Region-Growing

- 1) Find all connected components in $S(x,y)$ and reduce each connected component to one pixel. Label all such pixels found as 1. All other pixels in S are labeled 0.
- 2) Form an image f_0 such that, at each point (x,y) , $f_0(x,y) = 1$ if the input image satisfies a given predicate, P , at those coordinates, and $f_0(x,y) = 0$ otherwise.

Region Growing and Region Splitting

- 3) Let g be an image formed by appending to each seed point in S all the 1-valued points in f that are 8-connected to that seed point.
- 4) Label each connected component in g with a different region label (Ex. integers or letters). This is the segmented image obtained by region growing.

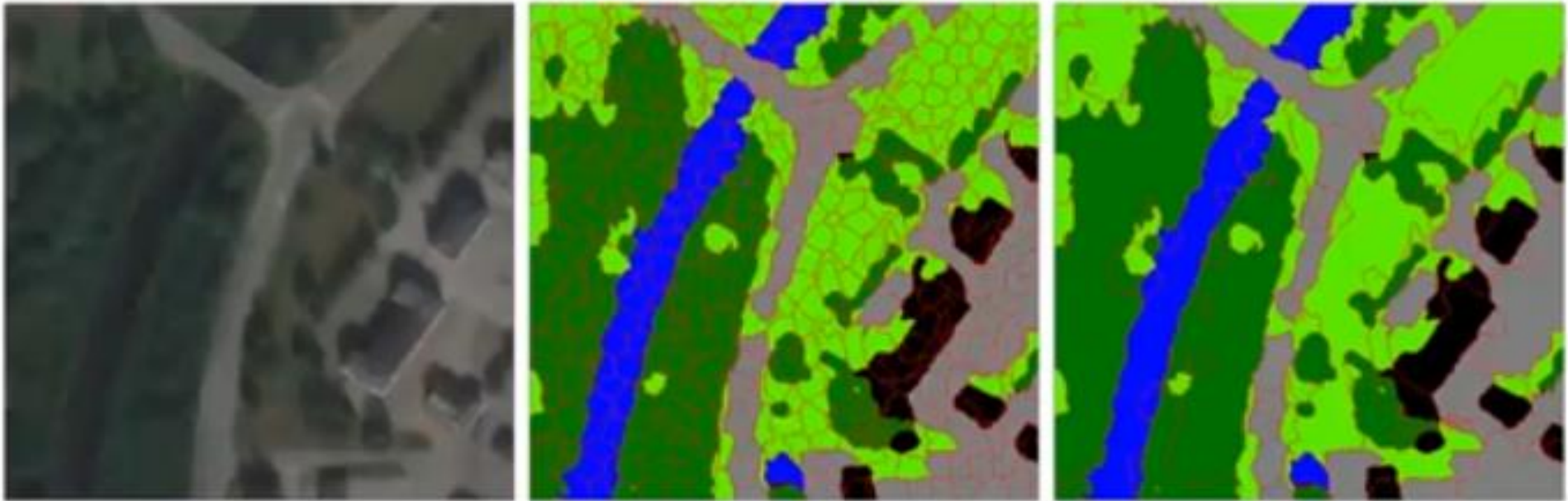
Region Growing and Region Splitting

Procedure for Region Splitting and Merging

- 1) If a region R is inhomogeneous ($P(R) = \text{False}$), then R is split into four sub-regions.
 - 2) If 2 adjacent regions R_i, R_j are homogeneous ($P(R_i \cup R_j) = \text{True}$), then they are merged.
 - 3) The algorithm stops when no further splitting or merging is possible.
- # Note: Condition for region growing:
 $\text{abs}(\text{seed value} - \text{pixel value}) \leq \text{Threshold}$

Region Merging/Growing

Used to segment an image into regions of similar intensity or color.
The algorithm is used to evaluate the values within a regional span and grouped together based on the **merging** criteria



Region Growing

Q1. Apply region growing on the following image with initial point at $(2,2)$ and threshold value as 2. Use 4 connectivity.

	0	1	2	3
0	0	1	2	0
1	2	5	6	1
2	1	4	7	3
3	0	2	5	1

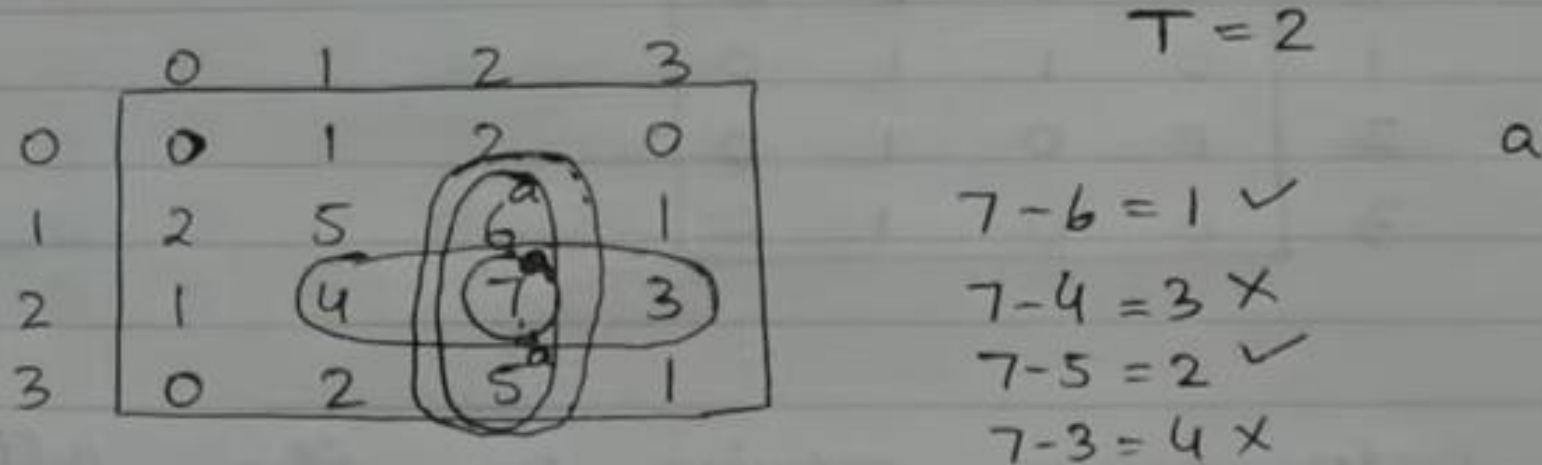
$T = 2$

Ans. The segmented region is shown in the following figure.

Condition $\rightarrow$ absolute difference ≤ 2 .
4 way connectivity

Region Growing

Q1. Apply region growing on the following image with initial point at (2,2) and threshold value as 2. Use 4 connectivity.

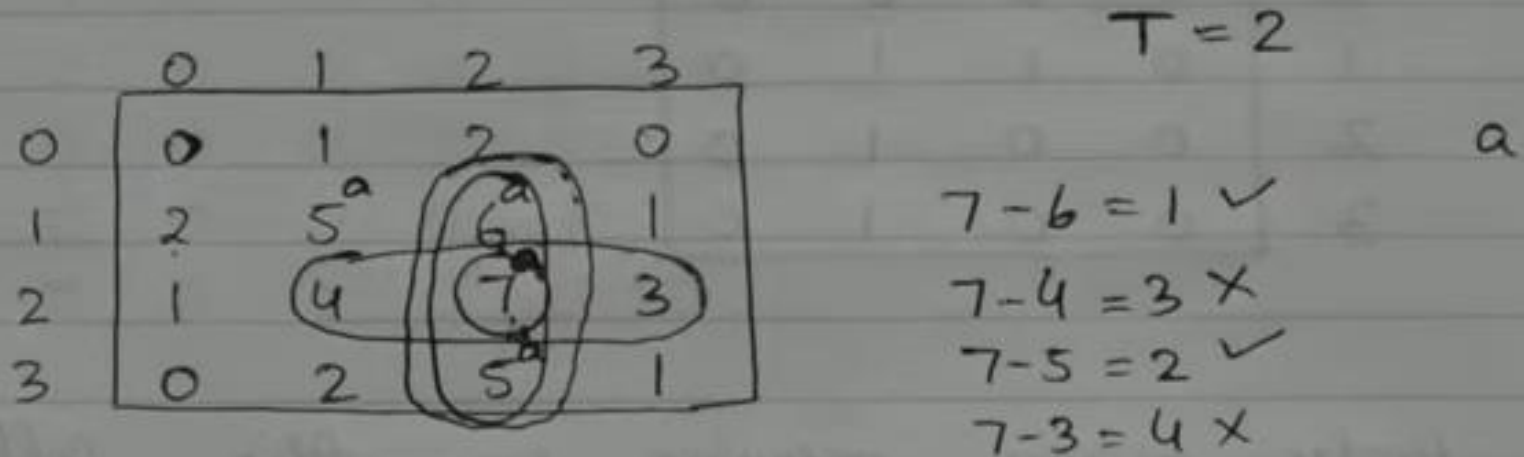


Ans. The segmented region is shown in the following figure.

Condition → absolute difference ≤ 2
4 way connectivity

Region Growing

Q1. Apply region growing on the following image with initial point at (2,2) and threshold value as 2. Use 4 connectivity.



Ans. The segmented region is shown in the following figure.

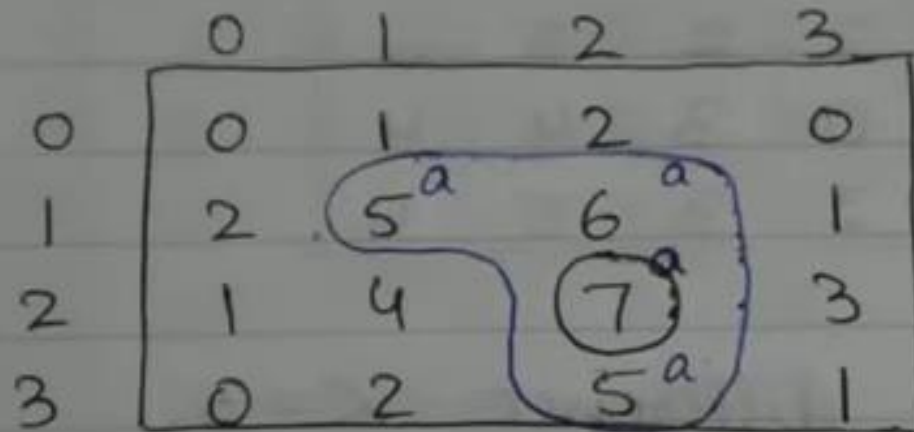
Condition → absolute difference ≤ 2 .
4 way connectivity

Region Growing

Ans. The segmented region is shown in the following figure.

Condition $\rightarrow$ absolute difference ≤ 2 .
4 way connectivity

Here, we will compare each element which is 4 way connected to the seed element.



Region Growing

Segmented image obtained from region growing:

	0	1	2	3
0	0	0	0	0
1	0	1	1	0
2	0	0	1	0
3	0	0	1	0

Region Growing

Q2. Apply region growing on the following image with seed point as 6 and threshold value as 3.

5	⑥	6	7	6	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Seed point = 6
 $T = 3$

Condition $\rightarrow$ absolute difference ≤ 3 .
8 way connectivity.

Region Growing

Q2. Apply region growing on the following image with seed point as 6 and threshold value as 3.

5 ^a	6 ^a	6 ^a	7 ^a	6 ^a	7 ^a	6 ^a	6 ^a
6 ^a	7 ^a	6 ^a	7 ^a	5 ^a	5 ^a	4 ^a	7 ^a
6 ^a	6 ^a	4 ^a	4 ^a	3	2	5 ^a	6 ^a
5 ^a	4 ^a	5 ^a	4	2	3	4 ^a	6 ^a
0	3	2	3	3	2	4 ^a	7 ^a
0	0	0	0	2	2	5 ^a	6 ^a
1	1	0	1	0	3	4 ^a	4
1	0	1	0	2	3	5	4

Seed point = 6
T = 3

a

Condition $\rightarrow$ absolute difference ≤ 3 .
8 way connectivity.

Region Growing

Q2. Apply region growing on the following image with seed point as 6 and threshold value as 3.

5 ^a	6 ^a	6 ^a	7 ^a	6 ^a	7 ^a	6 ^a	6 ^a
6 ^a	7 ^a	6 ^a	7 ^a	5 ^a	5 ^a	4 ^a	7 ^a
6 ^a	6 ^a	4 ^a	4 ^a	3	2	5 ^a	6 ^a
5 ^a	4 ^a	5 ^a	4	2	3	4 ^a	6 ^a
0	3	2	3	3	2	4 ^a	7 ^a
0	0	0	0	2	2	5 ^a	6 ^a
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Seed point = 6
T = 3

a

Condition $\rightarrow$ absolute difference ≤ 3 .
8 way connectivity.

Region Growing

1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	0	0	1	1
1	1	1	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	0	0

→ Segmented Image

Region Growing

Region Growing Segmentation

Example: Consider a 3-bit image given below. Assuming seed value of 6, segment the following image using 8-connectivity and threshold=3

How many segments are identified ?

Ans: Conditions for growing a region

8-Connectivity between seed pixel and given pixel

$\text{abs}(\text{seed_value} - \text{pixel_value}) < 3$

seed= 6

2	2	7	2	1
1	7	6	6	2
7	6	6	5	7
2	4	5	4	2
1	2	5	1	1



Applying thresholding condition

0	0	1	0	0
0	1	1	1	0
1	1	1	1	1
0	1	1	1	0
0	0	1	0	0



Region Growing

Aplying thresholding condition				
0	0	1	0	0
0	1	1	1	0
1	1	1	1	1
0	1	1	1	0
0	0	1	0	0

Aplying 8-connectivity to label each pixel				
a	a	b	c	c
a	b	b	b	c
b	b	b	b	b
d	b	b	b	e
d	d	b	e	e

Region Splitting

Q. Apply the region splitting on the following image
Assume the threshold value be ≤ 4

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Region Splitting

$$7 - 0 = 7$$

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

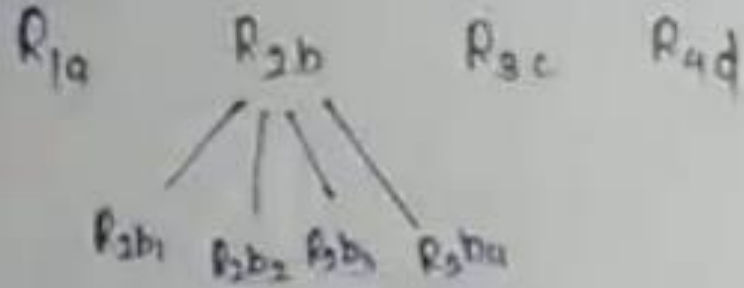
Region Splitting

R_{1a}
 R_{2b}
 R_{3c}
 R_{4d}

R_{1a}				R_{2b}			
5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4
R_{3d}				R_{3c}			

$7-0=7$
 $7-4=3$

Region Splitting



Q1a				Q2a		Q2b	
5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Q3c

$$7 - 0 = 7$$

$$7 - 4 = 3$$

$$7 - 2 = 5$$

$$7 - 5 = 2$$

$$7 - 4 = 3$$

$$6 - 4 = 2$$

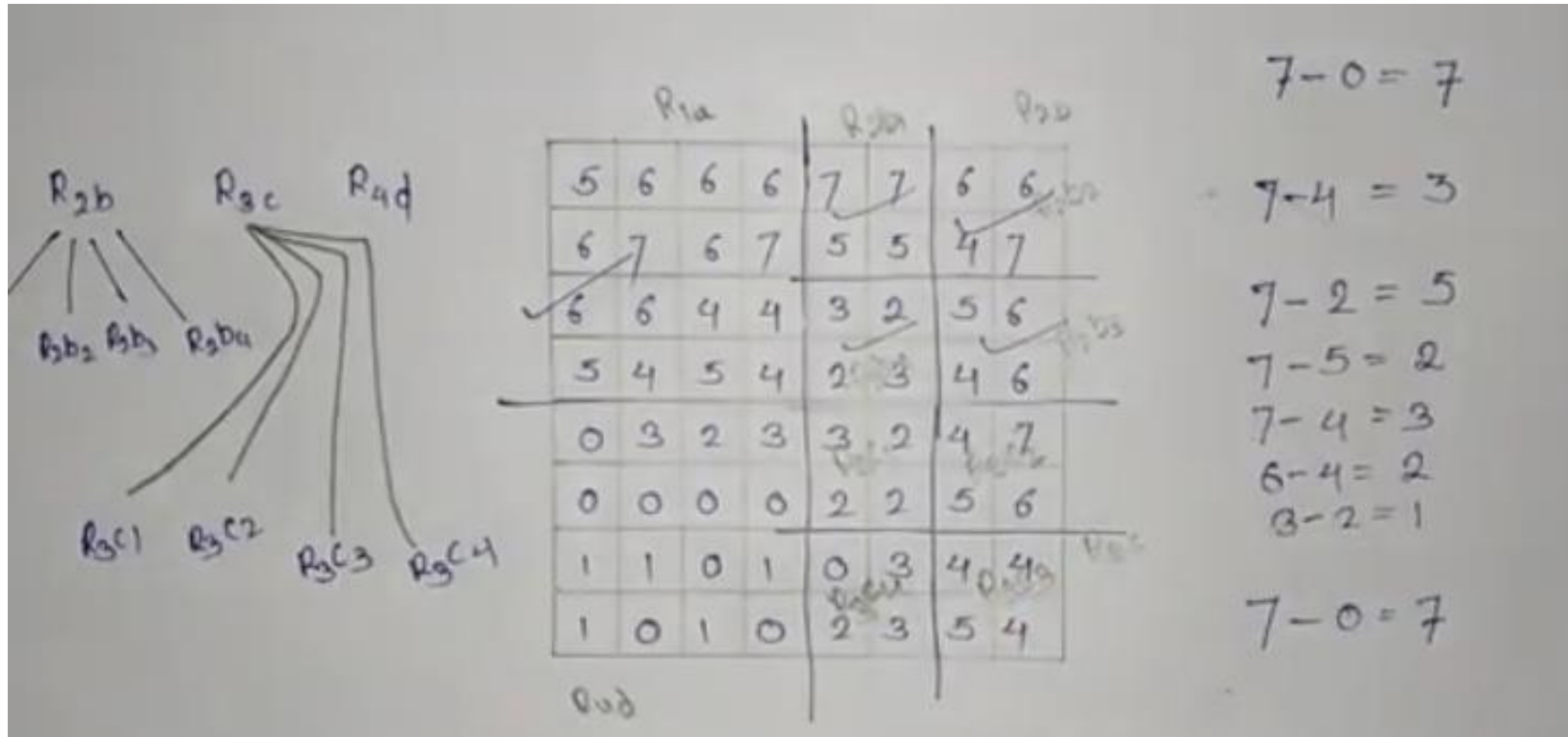
$$3 - 2 = 1$$

Region Splitting

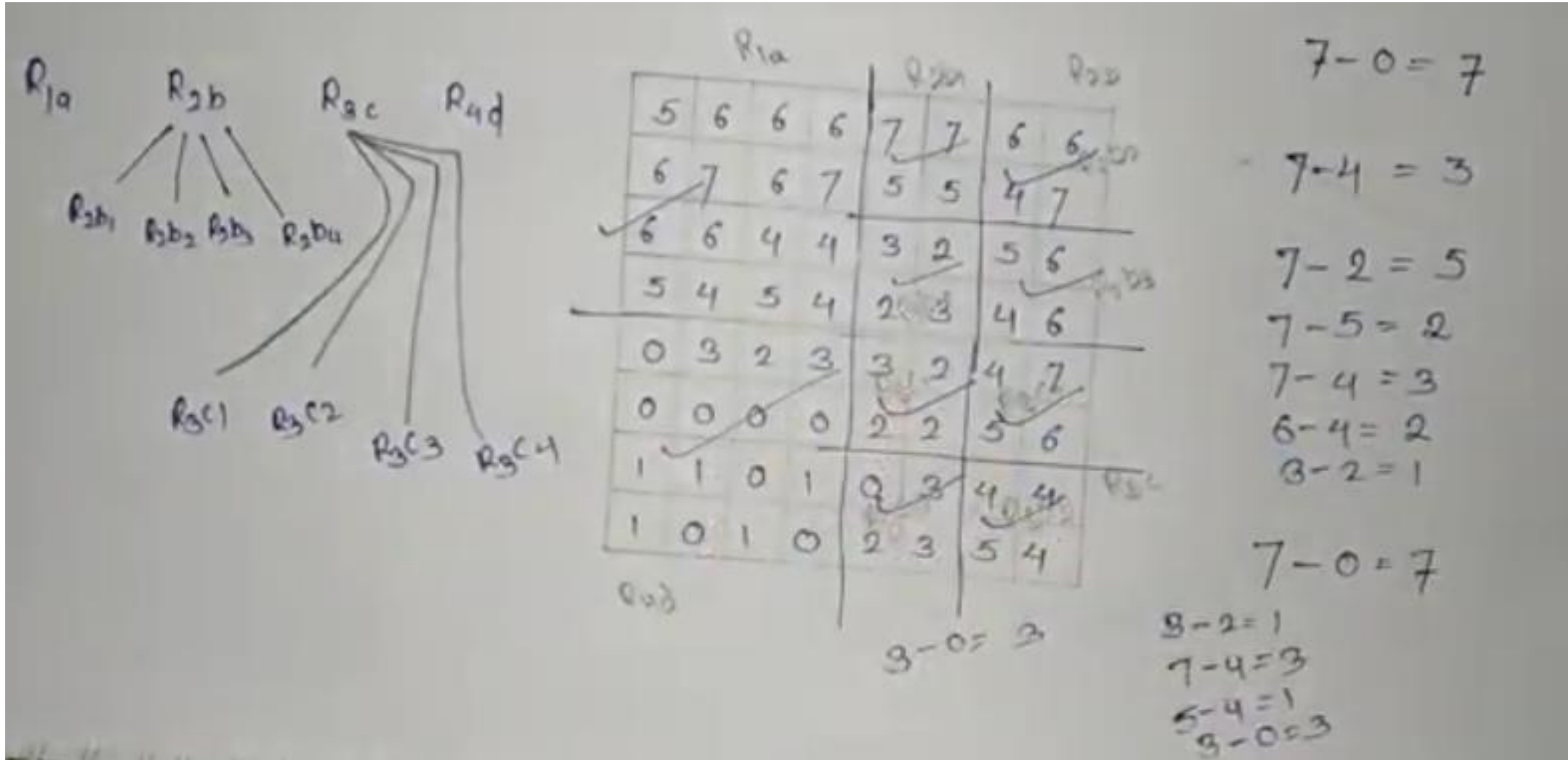
P1a				P2a		P2b	
5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4
Qud							

$7-0=7$
 $7-4=3$
 $7-2=5$
 $7-5=2$
 $7-4=3$
 $6-4=2$
 $3-2=1$
 $7-0=7$

Region Splitting



Region Splitting



Region splitting and Merging

Q3. Apply splitting and merging on the following image with threshold value equal to 3.

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Ans. Condition : absolute difference ≤ 3 .

Max value = 7

Min value = 0

$|7 - 0| = 7$ which is greater than 3.

Therefore we will split the region into 4 sub-regions.

Region splitting and Merging

Splitting

	5	6	6	6		7	7	6	6	
Ⓐ	6	7	6	7		5	5	4	7	Ⓑ
	6	6	4	4		3	2	5	6	
	5	4	5	4		2	3	4	6	
	0	3	2	3		3	2	5	7	
Ⓒ	0	0	0	0		2	2	5	6	Ⓓ
	1	1	0	1		0	3	4	4	
	1	0	1	0		2	3	5	4	

In region Ⓐ :

$$\text{Max} = 7$$

$$\text{Min} = 4$$

$$|7 - 4| = 3 \text{ which is equal to threshold.}$$

Therefore, no need to split.

Region splitting and Merging

Splitting

(a)	5	6	6	6	7	7	6	6	(b)	$7 - 2 = 5$
	6	7	6	7	5	5	4	7		
	6	6	4	4	3	2	5	6		
	5	4	5	4	2	3	4	6		
					3	2	5	7		
(c)	0	3	2	3	2	2	5	6	(d)	
	0	0	0	0	0	3	4	4		
	1	1	0	1	2	3	5	4		
	1	0	1	0						

Region splitting and Merging

<u>Splitting</u>							
(a)	5	6	6	6	7	7	6 6
	6	7	6	7	5	5	4 7
	6	6	4	4	3	2	5 6
	5	4	5	4	2	3	4 6
(c)	0	3	2	3	3	2	5 7
	0	0	0	0	2	2	5 6
	1	1	0	1	0	3	4 4
	1	0	1	0	2	3	5 4

(b)

$|7 - 2| = 5$

$|3 - 0| = 3$

(d)

$|7 - 0| = 7$

Region splitting and Merging

In region (b):

$$\text{Max} = 7 \quad \text{Min} = 2$$

$$|7 - 2| = 5 \text{ which is greater than } 3.$$

Therefore we will split the region (b) into 4 sub-regions.

In region (c):

$$\text{Max} = 3 \quad \text{Min} = 0$$

$$|3 - 0| = 3.$$

So no need to split.

In region (d):

$$\text{Max} = 7 \quad \text{Min} = 0$$

$$|7 - 0| = 7.$$

So we will split into 4 sub-regions.

Region splitting and Merging

					(b1)		(b2)		
	5	6	6	6	7	7	6	6	
(a)	6	7	6	7	5	5	4	7	(b)
	6	6	4	4	3	(b3) 2	5	(b4) 6	
	5	4	5	4	2	3	4	6	
	0	3	2	3	3	(d1) 2	5	(d2) 7	
(c)	0	0	0	0	2	2	5	6	(d)
	1	1	0	1	0	(d3) 3	4	(d4) 4	
	1	0	1	0	2	3	5	4	

Furthermore, we will check all the sub-regions. Since all of them are ≤ 3 , no further splitting is required.

Region splitting and Merging

Merging

Check adjacent regions, if they are falling within the threshold, then merge.

Consider regions (a) and (b)

$$\text{Max} = 7 \quad \text{Min} = 4$$

$$|7 - 4| = 3 \quad \checkmark \text{ Merge.}$$

Region splitting and Merging

(a)	5	6	6	6	7	7	6	6	(b)
	6	7	6	7	5	5	4	7	
	6	6	4	4	3	2	5	6	
	5	4	5	4	2	3	4	6	
	0	3	2	3	3	2	5	7	
(c)	0	0	0	0	2	2	5	6	(d)
	1	1	0	1	0	3	4	4	
	1	0	1	0	2	3	5	4	

Furthermore, we will check all the sub-regions. Since all of them are ≤ 3 , no further splitting is required.

Merging

Check adjacent regions, if they are falling within the threshold, then merge.

Consider regions (a) and (b)

Max = 7 Min = 4

 $|7-4| = 3 \quad \checkmark \text{ Merge.}$

Region splitting and Merging

Consider regions $(a\ b1)$ and $(b2)$

$$\text{Max} = 7 \quad \text{Min} = 4$$

$$|7 - 4| = 3 \quad \checkmark \text{ Merge.}$$

Consider regions $(a\ b1\ b2)$ and $(b4)$

$$\text{Max} = 7 \quad \text{Min} = 4$$

$$|7 - 4| = 3 \quad \checkmark \text{ Merge}$$

Consider regions $(a\ b1\ b2\ b4)$ and $d2$

$$\text{Max} = 7 \quad \text{Min} = 4$$

$$|7 - 4| = 3 \quad \checkmark \text{ Merge.}$$

Similarly, merge $(a\ b1\ b2\ b4\ d2)$ with $(d4)$.

Region splitting and Merging

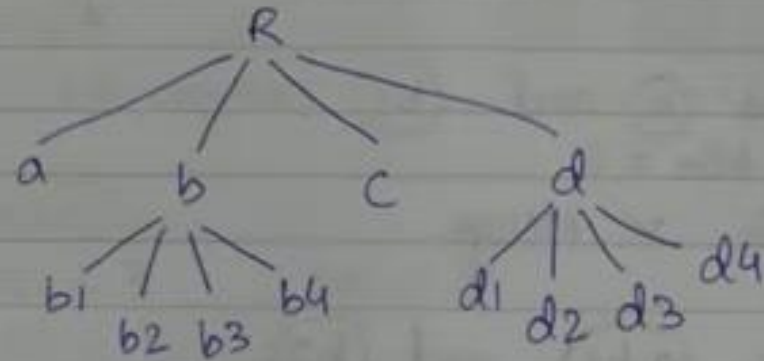
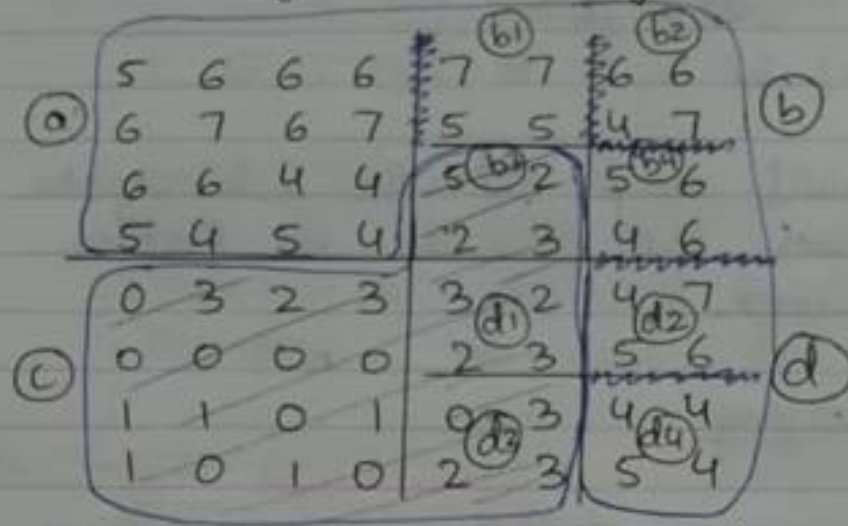
Similarly, merge (c), (d1), (b3) and (d3).

Final segmented image :

				(b1)	(b2)		
(a)	5	6	6	6	7	7	6 6
	6	7	6	7	5	5	4 7
	6	6	4	4	5 (b3)	2	5 (b4) 6
	5	4	5	4	2	3	4 6
(c)	0	3	2	3	3 (d1)	2	4 (d2) 7
	0	0	0	0	2	3	5 6
	1	1	0	1	0 (d3)	3	4 (d4) 4
	1	0	1	0	2	3	5 4
						(b)	(d)

Region splitting and Merging

Final segmented image :



Quadtree Structure
for Splitting

Thank you